Section 7.2 Span and Linear Independence

Let $V = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m\}$ be a nonempty set of vectors in a vector space V. The span of this set is denoted $span\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m\}$ and is defined to be the set of all linear combinations of the form

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_m\vec{v}_m$$

Where $c_1, c_2, \dots, c_m \in R$

If V consists of infinately many vectors, then we define $span(V)$ to be the set of all linear combinations of $finite$ subsets of V.

Let $V = \{x^2 + 1, 3x + 2, x^3\}$ determine if $x^3 + 2x^2 + 9x + 8 \in span(V)$

$\vec{v}_1 \quad \vec{v}_2 \quad \vec{v}_3$

$c_1(x^2+1) + c_2(3x+2) + c_3x^3 = x^3 + 2x^2 + 9x + 8$

$c_1x^2 + c_1 + c_2 3x + 2c_2 + c_3x^3$

$c_3x^3 + c_1x^2 + c_2 3x + (c_1 + 2c_2) = x^3 + 2x^2 + 9x + 8$

$\Rightarrow c_3 = 1 \quad c_1 = 2 \quad c_2(3) = 9 \quad c_1 + 2c_2 = 8$

$c_2 = 9/3 = 3 \quad 2 + 2(3) = 8 \Rightarrow 2 + 6 = 8 \checkmark$

Let $V = \left\{\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}\right\}$ Determine if $\begin{bmatrix} 5 & 0 \\ 6 & -2 \end{bmatrix} \in span(V)$ $\quad R^{2\times 2}$

$$c_1\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} + c_2\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 6 & -2 \end{bmatrix}$$

$$\begin{bmatrix} c_1 & 0 \\ 2c_1 & -c_1 \end{bmatrix} + \begin{bmatrix} 2c_2 & 0 \\ 0 & c_2 \end{bmatrix} = \begin{bmatrix} c_1+2c_2 & 0 \\ 2c_1 & -c_1+c_2 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 6 & -2 \end{bmatrix}$$

$c_1 + 2c_2 = 5$

$2c_1 = 6 \Rightarrow c_1 = 6/2 = 3$

$-c_1 + c_2 = -2 \Rightarrow -3 + 1 \stackrel{?}{=} -2 \checkmark$

$3 + 2c_2 = 5$

$2c_2 = 2$

$c_2 = 1$

Section 7.2 Span and Linear Independence

Consider the infinite vector space $V = \{1, x, x^2, x^3 \ldots\}$ Given the following equality,

$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \cdots = \sum_{n=0}^{\infty} x^n/n!$$

Is e^x in the $span\{V\}$?

No because we need only many vectors in the L.C.

Is $x^{6000} - 2x^{500} + 2x + 1$?

$c_1 x^{6000} + c_2 x^{500} + c_3 x + c_4(1)$ yes!

$c_1 = 1$, $c_2 = -2$, $c_3 = 2$, $c_4 = 1$

Suppose that V is a (possibly infinite) subset of a vector space V, and let $S = span(V)$. Then S is a subspace of V.

Proof:

Let $\vec{v}_1, \vec{v}_2 \in span\{V\}$. By the definition of span, $\vec{v}_1 + \vec{v}_2 \in span\{V\}$ and $r\vec{v}_1 \in V$. Also note that the linear combination $\vec{v}_1 + (-1)\vec{v}_1 = \vec{0} \in$ span(V).

$V = \{1, x, x^2, x^3, \ldots\}$

$S = span\{1, x, x^2\}$

Section 7.2 Span and Linear Independence

Let $\hat{V} = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m\}$ be a set of vectors in a vector space V. Then $\hat{V}$ is linearly independent if the equation

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_m\vec{v}_m = \vec{0}$$

has only the trivial solution $c_1 = \cdots = c_m = 0$ the set $\hat{V}$ is linearly dependent if there are any nontrival solutions.

Consider the vector space $V = \{1, x, x^2, x^3, \dots, x^n\}$ is V linearly independent?

Let's think about it, consider

$$c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0(1) = 0$$

Suppose that any of the $c_i's$ are nonzero, will the equation be zero for all possible x values ?

Example: Determine if $\{1, x, e^x\}$ are linearly independent. The trick here is to realize that if there are nonzero coefficients of a linear combonitation of the vectors that gives us zero, then it must work for all x values. We can therefore choose specific values and see if, for those values, all of the coefficients must be zero to get zero for all chosen $x's$. If so, we immeadiately know there is no choice of non zero coefficients for which we will always get a sum adding to zero that works for every x. Let's choose $x = 0, x = 1, x = 2$

$$c_1(1) + c_2 x + c_3 e^x = 0$$

C(R)

$e \approx 2.71828\ 1828\ 4590\ 45\dots$

$x = 0 \quad c_1 + c_2(0) + c_3 e^0 = 0$

$x = 1 \quad c_1 + c_2(1) + c_3 e = 0$

$x = 2 \quad c_1 + c_2(2) + c_3 e^2 = 0$

$c_1 + 0c_2 + c_3(1) = 0$

$c_1 + c_2 + c_3 e = 0$

$c_1 + 2c_2 + e^2 c_3 = 0$

```
octave:2> A=[1 0 1 0; 1 1 exp(1) 0; 1 2 exp(2) 0]
A =

   1.00000   0.00000   1.00000   0.00000
   1.00000   1.00000   2.71828   0.00000
   1.00000   2.00000   7.38906   0.00000

octave:3> rref(A)
ans =

   1   0   0   0
   0   1   0   0
   0   0   1  -0
```

Section 7.2 Span and Linear Independence

The set $V = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m\}$ of nonzero vectors is linearly dependent if and only if at least one vector in the set is in the span of the others.

Determine if $\{1, \cos(x), \cos(2x), \sin^2(x)\}$ is a linearly independent subset of $C(R)$.

$$\frac{1-\cos(2x)}{2} = \sin^2(x)$$

$$\frac{1}{2}(1) - \frac{1}{2}\cos(2x) = \sin^2(x)$$

$$c_1(1) + c_2\cos(2x) = \sin^2(x)$$

Linearly dependent

recall

$$\int \sin^2(x)\,dx$$

$$\int \frac{1-\cos(2x)}{2}$$

$$= \int \frac{1}{2} - \frac{1}{2}\int \cos(2x)$$

Let $\hat{V} = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m\}$ be a subset of a vector space V then:

(a) The set $\hat{V}$ is linearly independent if and only if the equation $c_1\vec{v}_1 + \cdots + c_m\vec{v}_m = \vec{v}$ has **at most** one solution for each $\vec{v} \in V$

(b) The set $\hat{V}$ spans V if and only if the equation $c_1\vec{v}_1 + \cdots + c_m\vec{v}_m = \vec{v}$ *has* ***at least*** *one solution for each* $\vec{v} \in V$

What might be a set of vectors that span $R^{2\times 2}$?

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in R^{2\times 2} \longrightarrow a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\}$$